From Drawing Product to Drawing Process: Combining 2D Sketch Analysis with Video-Based 3D Hand Kinematics for Parkinson's Motor Assessment

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***Abstract*—Parkinson's disease (PD) affects over 10 million people worldwide, with motor symptoms commonly assessed through drawing and handwriting tasks. Current digital assessment requires specialized tablets costing $500–$2000, limiting at-home monitoring. We present a two-part feasibility study combining 2D drawing product analysis with a novel 3D video-based hand kinematic process analysis for PD motor assessment.**

In Part A, using a public dataset of 204 drawings (102 PD patients, 102 healthy controls), we demonstrate that computational sketch features—including image entropy ($p = 0.0003$), stroke density ($p = 0.009$), and crossing count ($p = 0.001$)—significantly distinguish PD from healthy drawings. Six of 16 proposed features achieve statistical significance, and SVM classification reaches 69.7% accuracy on wave drawings.

In Part B, we propose a novel approach using smartphone video-based 3D hand tracking (MediaPipe) to analyze hand kinematics *during* the drawing act. In a proof-of-concept study with 2 healthy participants performing spiral tracing and sentence writing under 5 simulated PD conditions (normal, mild tremor, strong tremor, bradykinesia, micrographia), we observe preliminary, hypothesis-generating evidence — not yet confirmatory given the small n — that: (1) tremor-related kinematic features showed mixed but substantial changes under simulated tremor in this pilot, restricted to spiral-task recordings matched across both participants (normalized jerk: $+124.3\%$, tremor amplitude: $+240.7\%$, wrist range of motion: $+153.8\%$, peak speed: $+127.4\%$, speed variability: $+58.1\%$, PD-band power: $+32.0\%$); (2) simulated micrographia may be distinguishable from hand kinematics alone, with spectral arc length smoothness showing a statistically significant difference in this small cohort ($p = 0.019$, Cohen's $d = 3.2$); and (3) the 3D process approach appears to capture motor information not visible to 2D product analysis. These Part B findings motivate, but do not yet establish, the value of process analysis; they should be read as pilot evidence pending replication with a properly powered sample.

This combined framework bridges validated sketch product analysis (hardware-dependent) with novel sketch process analysis (requiring only a standard smartphone camera), establishing technical feasibility — pending clinical validation — for accessible PD motor monitoring.

***Index Terms*—Parkinson's disease, hand tracking, sketch analysis, motor assessment, tremor detection, MediaPipe, smartphone sensing, micrographia, SPARC, rehabilitation monitoring**

I. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder, affecting over 10 million people worldwide [1]. The cardinal motor symptoms—resting tremor (4–6 Hz), bradykinesia, rigidity, and postural instability—progressively impair fine motor control. Handwriting and drawing tasks are among the earliest and most sensitive clinical indicators of motor decline [2].

Clinical assessment relies on the Unified Parkinson's Disease Rating Scale (UPDRS) [3], administered during clinic visits every 2–4 months. Between visits, patients lack objective motor monitoring, making medication adjustment reactive. Drawing tasks—particularly the Archimedes spiral test and handwriting analysis for micrographia—are standard clinical tools [4], yet their scoring remains largely subjective or requires specialized tablets costing $500–$2000 [5].

Two fundamentally different approaches exist for computational drawing analysis: (1) *product analysis*—examining the final drawn output, and (2) *process analysis*—examining how the drawing was produced. Product analysis has been extensively studied [5], [6]. Process analysis, which captures richer temporal and kinematic information, has been limited to specialized hardware. Recent advances in markerless hand tracking— particularly Google's MediaPipe framework [7], which provides real-time 3D estimation of 21 hand joint positions from monocular RGB video—now enable process analysis using only a standard smartphone camera.

Research Gap. While 2D drawing product analysis for PD is well-established, to the best of our knowledge, no prior study has investigated: (1) whether video-based 3D hand kinematics during drawing can detect PD motor symptoms, or (2) how product and process analysis complement each other for comprehensive motor assessment.

Contributions. This paper presents:

1) A two-part framework combining 2D drawing product analysis (validated on 204 real PD/healthy drawings) with novel 3D video-based hand kinematic process analysis.

2) A 16-feature motor assessment framework for each modality—Part A features derived from drawing images; Part B features from 3D hand joint trajectories.
3) Preliminary evidence, from an $n = 2$ pilot cohort, that video-based hand kinematics respond with large effect sizes to simulated PD symptoms, and that micrographia may be distinguishable from hand movement alone without analyzing written text ($p = 0.019$, Cohen's $d = 3.2$) — a hypothesis-generating result that motivates, rather than establishes, this direction.
4) A discussion of how process analysis could in principle capture motor information invisible to product analysis, enabling future accessible, hardware-free PD monitoring, contingent on clinical-scale validation.

II. Related Work

A. Drawing and Handwriting Analysis for PD

Drawing tasks have been used for PD motor assessment for decades [4]. The Archimedes spiral test is widely used to quantify tremor severity, with features such as drawing speed, smoothness, and deviation from template correlating with UPDRS motor scores [15]. Micrographia—progressive reduction in handwriting size—affects 50–60% of PD patients and often precedes clinical diagnosis [16].

Digital tablet-based systems have enabled quantitative analysis of pen dynamics including pressure, velocity, and jerk [5], [6]. Drotar et al. [5] demonstrated that kinematic features from digitized handwriting could distinguish PD patients from controls with over 80% accuracy, but these systems require specialized hardware unavailable in home settings.

B. Sketch Recognition and Computational Drawing Analysis

Computational analysis of the drawing process has a rich history in the sketch recognition community. Early work by Rubine [8] established gesture feature extraction from pen trajectories, and Gross [9] examined the ambiguity of hand-drawn marks as a fundamental property of freehand input. Hammond and Davis [11] introduced the LADDER language for formally describing and recognizing hand-drawn sketch primitives, enabling systematic analysis of drawing structure. Paulson and Hammond [12] advanced geometric sketch recognition with accurate primitive recognition and beautification, while Fonseca et al. [10] demonstrated robust sketch understanding from noisy hand-drawn input.

A particularly relevant line of work examines not just the sketch product but the drawing process as a signal of cognitive and motor ability. Hammond et al. [13] proposed metrics for modeling expert sketching ability that capture fluency, temporal dynamics, and process characteristics—features conceptually parallel to our motor health assessment approach. Separately, rehabilitation systems such as Kinohaptics [14] have shown that continuous motor monitoring with automated haptic feedback significantly improves post-surgical recovery, motivating similar approaches for neurodegenerative conditions.

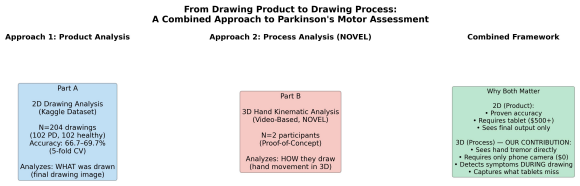


Fig. 1. Combined framework: Part A analyzes the 2D drawing product using image features (validated on real PD patients). Part B analyzes the 3D hand drawing process using video-based kinematics (novel; requiring only a smartphone camera).

C. Video-Based Hand Analysis

MediaPipe Hands [7] provides real-time 3D estimation of 21 hand landmarks from monocular video. While extensively applied to gesture recognition and sign language [17], its use for clinical motor assessment remains largely unexplored. Yan et al. [18] recently applied MediaPipe to hand tremor quantification. However, to the best of our knowledge, no prior study has combined video-based hand tracking with sketch-process analysis for PD motor symptom detection during functional drawing tasks.

III. Methodology

Our framework consists of two complementary approaches (Fig. 1): Part A analyzes the drawing *product* (what was drawn) using image processing on a public clinical dataset, while Part B analyzes the drawing *process* (how it was drawn) using video-based 3D hand tracking.

A. Part A: 2D Drawing Product Analysis

1) Dataset: We use the publicly available Parkinson's Drawing Dataset [23], containing 204 drawings from 102 PD patients and 102 age-matched healthy controls. Each participant produced both spiral and wave drawings (102 per type). All drawings were scanned at consistent resolution.

2) Feature Extraction: We propose 16 image-based features:

- **Tremor features**: Contour smoothness (curvature variability), tremor intensity (high-frequency spectral power), stroke irregularity (deviation from smooth fit).
- **Kinematic proxy features**: Stroke density, speed proxy (inter-point distance), speed variability.
- **Shape features**: Drawing area, compactness, radial deviation, crossing count, Hu moments (1st and 2nd).
- **Texture features**: Stroke thickness variation, mean thickness, perimeter-area ratio, image entropy.

3) Classification: Three classifiers were evaluated using stratified 5-fold cross-validation: Random Forest (100 trees, max depth 10), Gradient Boosting (100 estimators, max depth 5), and SVM with RBF kernel ($C = 1.0$, $\gamma = \texttt{scale}$). All features were z-score normalized.

B. Part B: 3D Video-Based Hand Kinematic Analysis

1) Participants and Setup: Two healthy participants (both male, ages 30 and 60) provided informal consent for this proof-of-concept study. A consumer smartphone (rear camera, 1080p

at 30–60 fps) was mounted 20–25 cm above the drawing surface. Participants drew on white A4 paper under consistent overhead lighting. No specialized hardware was used. We emphasize that with $n = 2$, this study is designed to establish technical and methodological feasibility, not to provide statistically generalizable estimates of effect size.

2) Drawing Tasks:

- **Spiral tracing**: Tracing a printed Archimedes spiral template (3 loops, 10 cm diameter).
- **Sentence writing**: Writing "Today is a beautiful day" on blank paper.

3) Simulated Symptom Conditions: To provide grounded kinematic reference ranges prior to clinical data collection, we used established physical simulation approaches [21]:

- **Normal**: No manipulation (baseline).
- **Mild tremor**: Vibrating motor device loosely secured near the drawing wrist to induce low-amplitude oscillation.
- **Strong tremor**: Same device firmly applied to increase oscillation amplitude.
- **Restricted-Movement Condition**: Thick elastic wrap around the wrist to restrict range of motion, intended as a kinematic perturbation related to reduced movement amplitude, not a physiological simulation of bradykinesia, which clinically involves slowness and amplitude decrement arising from distinct neuromotor mechanisms.
- **Micrographia**: Participant instructed to write with progressively decreasing letter size [16].

Each condition lasted approximately 20–50 seconds per task. Total recordings: 12 videos (2 participants $\times$ 6 conditions).

4) Hand Tracking Pipeline: Videos were processed using MediaPipe Hands [7], extracting 21 three-dimensional hand joint positions per frame. Detection and tracking confidence thresholds were set to 0.5. Mean detection rate across all videos was 63.97% (range: 10.7%–95.6%). Frames without a successful detection were not retained; only frames with valid landmarks were kept, each paired with its true elapsed timestamp. Velocity was computed as displacement divided by the actual (not assumed-uniform) time interval between consecutive retained frames, so detection gaps do not introduce artificial spikes into the derivative estimates. However, any rapid motion occurring within a detection gap is not captured and is instead averaged over the gap's true duration, which can understate peak velocity, jerk, or tremor content during low-detection-rate intervals. This is a limitation of the current proof-of-concept implementation, addressed further in Section 5.4.

5) Feature Extraction: We propose 16 hand kinematic features organized by PD symptom domain:

- **Bradykinesia**: Mean speed, speed variability, peak speed, normalized jerk [20], spectral arc length smoothness (SPARC) [19].
- **Tremor**: Tremor amplitude (3–12 Hz bandpass RMS), tremor frequency, PD-band power ratio (4–6 Hz).
- **Stability**: Grip stability (thumb-index distance variance), hand steadiness (wrist variability), wrist range of motion, finger coordination (inter-finger velocity correlation).

TABLE I
PART A: STATISTICALLY SIGNIFICANT FEATURES (PD VS. HEALTHY)

Feature	Healthy	PD	p	d
Image Entropy	5.159 ± 0.23	5.291 ± 0.29	0.0003***	0.51
Crossing Count	44.6 ± 53.8	93.7 ± 144.3	0.0015**	0.45
Speed Proxy	1.171 ± 0.02	1.177 ± 0.02	0.0062**	0.39
Stroke Density	0.061 ± 0.02	0.070 ± 0.03	0.0090**	0.37
Hu Moment 1	1.942 ± 0.15	2.003 ± 0.19	0.0109*	0.36
Perim.–Area Ratio	50.6 ± 15.1	45.1 ± 20.1	0.0300*	0.31

$*p < 0.05$, $**p < 0.01$, $***p < 0.001$. d: Cohen's d. Values: mean $\pm$ SD.

TABLE II
PART A: CLASSIFICATION ACCURACY (%, STRATIFIED 5-FOLD CV)

Task	RF	GB	SVM
Spiral	68.7 ± 7.0	62.8 ± 3.6	66.7 ± 5.7
Wave	68.6 ± 4.2	69.6 ± 5.5	$\mathbf{69.7 \pm 4.3}$
Combined	66.7 ± 6.2	—	—

RF: Random Forest. GB: Gradient Boosting. SVM with RBF kernel.

- **Temporal**: Drawing duration, pause ratio, velocity peaks per second, amplitude decay slope (micrographia index).

Let $\mathbf{p}_t \in \mathbb{R}^3$ denote the index fingertip position at frame t. The SPARC smoothness metric is computed as:

$$\text{SPARC} = -\int_0^{\omega_c} \sqrt{\left(\frac{1}{\omega_c}\right)^2 + \left(\frac{d\hat{V}(\omega)}{d\omega}\right)^2}\, d\omega \qquad (1)$$

where $\hat{V}(\omega)$ is the normalized Fourier magnitude spectrum of the velocity profile, and ω_c is the cutoff frequency (10 Hz) [19]. More negative SPARC values indicate smoother (healthier) movement.

IV. RESULTS

A. Part A: 2D Drawing Product Analysis

1) Statistical Analysis: Independent samples t-tests revealed 6 of 16 features significantly differing between PD and healthy groups ($p < 0.05$). Table I presents results for all significant features.

Image entropy was the most discriminative feature ($p = 0.0003$, Cohen's $d = 0.51$), with PD drawings exhibiting higher entropy (more irregular patterns). Crossing count was significantly elevated in PD ($p = 0.0015$), consistent with tremor-induced stroke crossings.

2) Classification Results: Table II summarizes classification accuracy.

Best single-task performance was achieved by SVM on wave drawings (69.7%). Fig. 2 shows the confusion matrix for the combined classifier.

3) Feature Importance: Fig. 3 shows the Random Forest feature importance ranking. Entropy, stroke density, and speed proxy were the top-3 discriminative features.

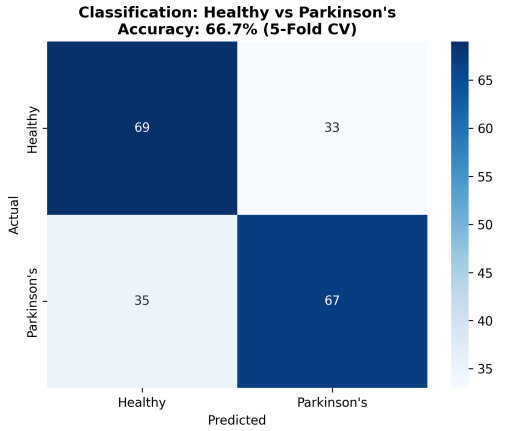


Fig. 2. Part A: Confusion matrix (5-fold CV, combined spiral and wave, Gradient Boosting, 66.7% accuracy).

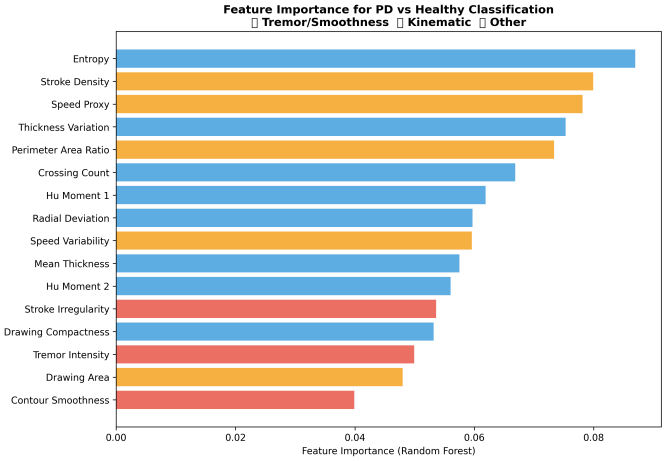


Fig. 3. Part A: Feature importance (Random Forest). Kinematic proxy features (orange) and structural features (blue) dominate.

B. Part B: 3D Video-Based Hand Kinematic Analysis

Note on interpretation: All results in this subsection derive from $n = 2$ participants and are reported as pilot, hypothesis-generating findings. Percentage changes and effect sizes below describe what was observed in this small cohort; given the sample size, large effect sizes are expected even for noisy or unreliable signals, and should not be read as precise or generalizable estimates.

1) Hand Tracking Performance: MediaPipe tracked the hand in 63.97% of frames on average (range: 10.7%–95.6%). Higher detection rates correlated with better lighting and reduced motion blur. Videos with detection rates $> 50\%$ (10 of 12) provided sufficient temporal resolution for reliable feature extraction.

2) Tremor Detection: Comparing the normal baseline with simulated mild tremor, restricted to spiral-task recordings and matched across both participants (2 recordings per condition), revealed a consistent increase across all six kinematic features in this pilot cohort (Fig. 4):

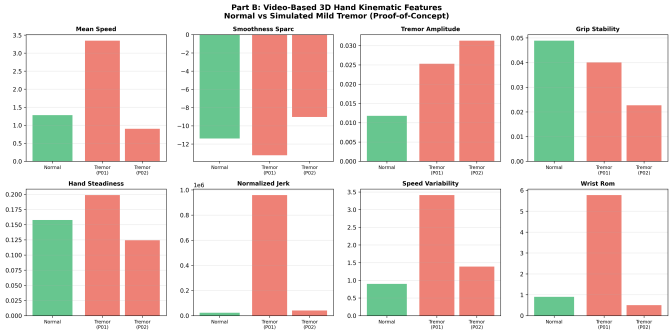


Fig. 4. Part B: 3D hand kinematic features comparing normal drawing (green) with simulated mild tremor (red) in the $n = 2$ pilot cohort. Differences were consistent in direction across both participants.

TABLE III
PART B: MICROGRAPHIA DETECTION (NORMAL VS. SIMULATED)

Feature	Normal	Micro.	p	d
SPARC Smoothness	-11.31 ± 0.31	-12.35 ± 0.34	0.019*	3.20
Duration (s)	22.12 ± 6.53	38.96 ± 3.88	0.031*	3.14

$*p < 0.05$. Effect sizes (Cohen's $d > 3$) reflect the very small n and should not be read as generalizable magnitudes. Results are proof-of-concept pending clinical validation.

- Tremor amplitude (3–12 Hz RMS): $+240.7\%$ increase.
- Wrist range of motion: $+153.8\%$ increase.
- Peak speed: $+127.4\%$ increase.
- Normalized jerk: $+124.3\%$ increase.
- Speed variability: $+58.1\%$ increase.
- PD-band power ratio (4–6 Hz): $+32.0\%$ increase.

All six features moved in the direction expected under a tremor-like perturbation. Given $n = 2$ recordings per condition, these percentage changes should be read as pilot-level, hypothesis-generating observations rather than precise or generalizable magnitude estimates; a properly powered follow-up is needed to establish stable effect sizes.

3) Micrographia Detection: Comparing normal writing ($n = 4$ recordings) with simulated micrographia ($n = 2$), independent samples t-tests revealed two statistically significant features (Table III). We note explicitly that these six recordings are repeated measurements from the same two participants, not independent observations from six different individuals; treating them as statistically independent in a t-test does not account for within-participant correlation and constitutes pseudoreplication. We report these tests for descriptive purposes only, not as confirmatory inferential statistics.

SPARC smoothness was more negative during micrographia ($p = 0.019$, Cohen's $d = 3.2$), indicating rougher movement in this pilot. Drawing duration was also longer ($p = 0.031$, Cohen's $d = 3.14$). These results are consistent with the hypothesis that micrographia could be detected from hand kinematics alone—without analyzing the written text—which would make detection language-agnostic and independent of text visibility (Fig. 5); confirming this requires a larger, diagnosed-patient sample.

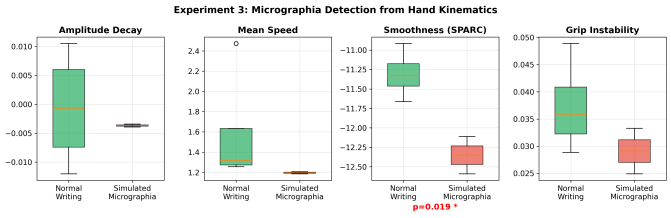


Fig. 5. Part B: In this $n = 2$ pilot, SPARC smoothness differed between normal writing and simulated micrographia ($p = 0.019$*), without requiring analysis of the written output. Pending replication at scale.

4) Restricted-Movement Condition Observation: Under the restricted-movement condition, tremor amplitude increased substantially (0.083 vs. 0.035 baseline) and smoothness degraded (SPARC: -8.79 vs. -11.31 baseline). Formal statistical testing was not feasible at $n = 2$ per condition; the direction and magnitude are consistent with expected PD motor signatures but are reported here as observational, not statistical, evidence, and this condition approximates reduced movement amplitude rather than the full clinical presentation of bradykinesia.

V. Discussion

A. Complementary Nature of Product and Process Analysis

Product and process analysis capture different aspects of PD motor impairment. **Product analysis (Part A)** captures the cumulative effect on the final drawing (69.7% accuracy, validated on 204 real drawings), but cannot detect tremor that was motor-compensated, or hesitations that did not alter the final stroke. **Process analysis (Part B)**, in principle, captures the moment-to-moment motor state during drawing—tremor, grip stability, smoothness, and velocity dynamics—potentially detecting motor phenomena before they manifest in the product. Our small pilot shows large relative changes in kinematic features under mild tremor, which is an encouraging but preliminary signal of sensitivity, not a validated finding. The combined framework's long-term promise is product analysis for validated large-scale screening paired with process analysis for daily monitoring at zero additional hardware cost — contingent on Part B being validated at a much larger scale.

B. Key Findings

Micrographia from hand kinematics (preliminary). The pilot result that SPARC smoothness differed between normal and simulated micrographia writing ($p = 0.019$, $n = 2$) is a hypothesis worth testing at scale, not yet a clinical finding. If replicated in diagnosed patients, language-agnostic detection independent of text visibility would be clinically meaningful. The idea that *how* a task is executed may carry diagnostic signal beyond *what* is produced [13] motivates extending this principle to other motor assessment tasks beyond drawing.

Feature sensitivity (preliminary). The relative changes observed in Part B (up to $+240.7\%$ for tremor amplitude in this $n = 2$ cohort) suggest 3D hand kinematics may be sensitive to simulated perturbation, though percentage-based effect estimates from $n = 2$ per condition remain imprecise and should not be read as stable effect sizes, in contrast to the smaller, more tightly estimated effect sizes in Part A (Cohen's d from 0.31 to 0.51 on $n = 204$). This comparison should be read as motivating further study, not as evidence that Part B is quantitatively more sensitive than Part A — the two were measured under very different sample sizes and conditions (simulated vs. real PD).

Accessibility. Because Part B uses only a consumer smartphone rather than specialized tablet hardware, the approach has the potential to reduce hardware barriers to resource-limited or home-based motor monitoring, if the underlying approach is validated.

C. Broader Implications

Our framing of PD motor assessment as a *sketch process analysis* problem opens a productive connection to HCI and sketch recognition research. The kinematic features we extract—smoothness, velocity profiles, temporal dynamics, and spatial amplitude—are structurally similar to the features used to analyze drawing fluency and motor expertise in freehand input systems [8], [13]. Process analysis in Part B is conceptually equivalent to studying *how* someone draws rather than *what* they drew, a distinction central to sketch recognition systems [10], [11]. This analogy suggests that feature engineering advances from the sketch recognition literature could be applied to clinical motor assessment—a cross-disciplinary direction that warrants dedicated, larger-scale future investigation.

D. Limitations

- **Part A accuracy**: 66.7–69.7% is moderate. Deep learning approaches may improve performance but were outside the scope of this feasibility study.
- **Part B sample size**: With $n = 2$ healthy participants, Part B is strictly proof-of-concept. The statistically significant micrographia result should be interpreted cautiously: at this sample size, large effect sizes are expected regardless of whether the underlying effect is real, and clinical validation with diagnosed patients is essential before any clinical claim can be made.
- **Simulated vs. real PD**: Physical symptom simulation in healthy participants does not fully replicate the neurological mechanisms of PD, and results may not transfer to real patients.
- **Detection variability and reduced temporal resolution during gaps**: Hand tracking ranged from 10.7% to 95.6% depending on lighting and motion blur. Strong tremor conditions had lower detection rates due to blur. Frames without a successful detection were dropped rather than interpolated; velocity was computed using the true elapsed time between retained frames, so gaps do not introduce artificial spikes, but any rapid motion within a gap is averaged over its true duration rather than captured, which can understate peak velocity, jerk, or tremor content during low-detection-rate intervals. A properly powered follow-up should implement explicit interpolation or uncertainty-

aware gap handling and test how this affects feature stability.

- **Repeated measurements, not independent samples**: The micrographia comparison (Section 4.2.3) uses six recordings drawn from only two participants. Treating these as six independent observations, as a standard t-test assumes, does not account for within-participant correlation (pseudoreplication); the reported p-value and effect size should be read as descriptive of this specific pilot, not as evidence from six independent individuals.
- **Cross-validation is not grouped by participant**: Because each of the 204 participants in Part A contributed both a spiral and a wave drawing (and in some cases multiple drawings per task), subject-level grouping during fold splitting is necessary to prevent one participant's drawings from appearing in both the training and validation sets. We confirm that the classification pipeline uses `StratifiedKFold`, which balances only the class label across folds and does not account for participant identity; no participant-level grouping (e.g., `StratifiedGroupKFold`) is implemented. The reported 66.7–69.7% accuracy figures should therefore be treated as an upper bound that may be inflated by participant-level leakage to an undetermined degree, pending a re-analysis with participant-grouped cross-validation.
- **Individual baseline variability**: The population-level thresholds used here assume a shared healthy baseline across individuals. In practice, resting kinematic smoothness and tremor likely vary meaningfully person-to-person, meaning a fixed, group-derived threshold may under- or over-detect deviation for any given individual. A personalized calibration approach, establishing each person's own baseline before flagging deviation from it, would likely be necessary for reliable individual-level monitoring, rather than population-wide norms alone.

E. Future Work

Immediate priorities: (1) clinical validation with diagnosed PD patients (Hoehn & Yahr I–IV) at a sample size sufficient for confirmatory statistical inference; (2) larger samples ($n > 30$ per condition); (3) longitudinal daily monitoring studies; (4) deep temporal models including state-space architectures [22] for hand trajectory classification; (5) unified product-plus-process feature fusion; (6) a real-time smartphone application providing clinician-interpretable motor monitoring output; and (7) extending the same kinematic feature pipeline from passive assessment toward interactive, immersive delivery—for example, virtual reality environments that gamify repeated drawing or writing exercises—which could in principle support motor rehabilitation rather than measurement alone, though this remains an unexplored direction requiring dedicated future work.

VI. Conclusion

We presented a two-part feasibility study of complementary approaches to Parkinson's motor assessment through drawing analysis. Part A validated that 16 image-based sketch features distinguish PD from healthy controls (best: 69.7% SVM accuracy, $n = 204$), with image entropy and stroke density as top discriminators — this is the more statistically robust component of the study. Part B, a small $n = 2$ pilot, found preliminary, hypothesis-generating evidence that novel 3D video-based hand kinematics—requiring only a consumer smartphone—respond consistently with simulated tremor across all six measured kinematic features (up to $+240.7\%$ for tremor amplitude in this pilot) and may distinguish simulated micrographia from hand movement alone ($p = 0.019$, Cohen's $d = 3.2$). These Part B findings motivate, but do not establish, the clinical value of process analysis.

The transition from drawing product to drawing process analysis represents a promising direction: from analyzing *what* patients drew (requiring specialized hardware) to analyzing *how* they draw (requiring only a phone camera). Future clinical validation with diagnosed PD patients, at a sample size adequate for confirmatory inference, will determine whether these proof-of-concept results generalize to real-world motor assessment and longitudinal monitoring.

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